



Oracle Database Administration

I

Chapter 10: Data Dictionary Fundamentals

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Objectives:

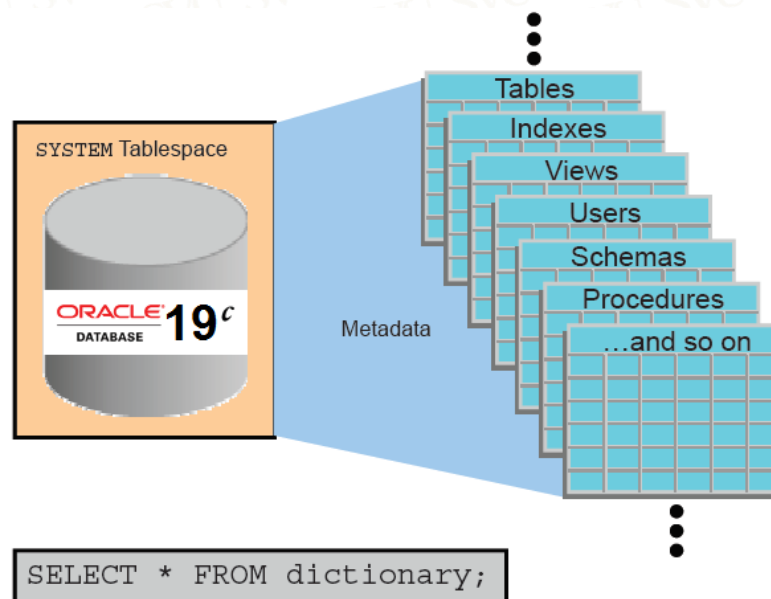
1. Data Dictionary Contents
2. Static Views
3. Dynamic Performance Views
4. Problem solving using data dictionary views

1. Data Dictionary Architecture:

The Oracle data dictionary is the metadata of the database and contains the names and attributes of all objects in the database.

The data dictionary:

- Is used by the Oracle Database server to find information about users, objects, constraints, and storage
- Is maintained by the Oracle Database server as object structures or definitions are modified
- Is available for use by any user to query information about the database
- Is owned by the SYS user
- Should never be modified directly using SQL



Oracle's data dictionary is vast and robust. Almost every conceivable piece of information about your database is available for retrieval. The data dictionary stores critical information about the physical characteristics of the database, users, objects, and dynamic performance metrics. A senior-level DBA must possess an expert knowledge of the data dictionary. Oracle provides two general categories of read-only data dictionary views:

- The contents of your database, such as users, tables, indexes, constraints, and privileges. These are sometimes referred to as the static `CDB/DBA/ALL/USER` data dictionary views, and they are based on internal tables stored in the `SYSTEM` tablespace. The term static, in this sense, means that the information within these views only changes as you make changes to your database, such as adding a user, creating a table, or modifying a column.
- A real-time view of activity in the database, such as users connected to the database, SQL currently executing, memory usage, locks, and I/O statistics. These views are based on virtual memory tables and are referred to as the dynamic performance views. The information in these views is continuously updated by Oracle as events take place within the database. The views are also sometimes called the `V$` or `GV$` views.

1.1. Static Views:

Oracle refers to a subset of the data dictionary views as static. These views are based on physical tables maintained internally by Oracle. These views are static in the sense that the data they contain do not change at a rapid rate (at least, not compared with the dynamic `V$` and `GV$` views). It is important to understand this architectural when querying the data dictionary. there are four levels that is applicable when using the container/pluggable database feature:

USER

The `USER` views contain information available to the current user. For example, the `USER_TABLES` view contains information about tables owned by the current user. No special privileges are required to select from the `USER-level` views.

ALL

The `ALL` views show you all object information the current user has access to. For example, the `ALL_TABLES` view displays all database tables on which the current user can perform any type of DML operation. No special privileges are required to query from the `ALL-level` views.

DBA

The DBA views contain metadata describing all objects in the database (regardless of ownership or access privilege). To access the DBA views, a DBA role or `SELECT_CATALOG_ROLE` must be granted to the current user.

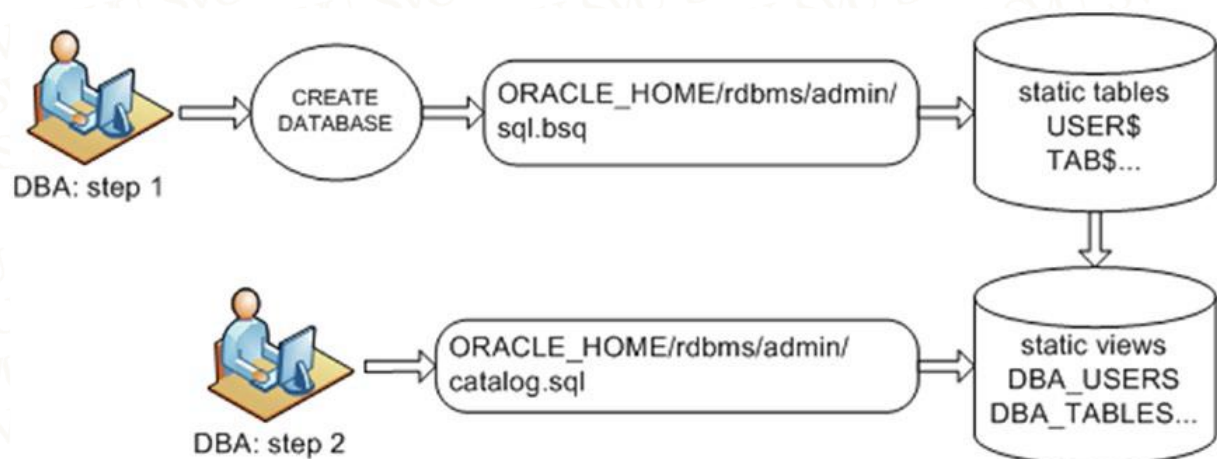
CDB

The CDB-level views are only applicable if you are using the pluggable database feature. This level provides information about all pluggable databases within a container database (hence the acronym CDB). You will notice that many of the static data dictionary and dynamic performance views have a new column, `CON_ID`. This column uniquely identifies each pluggable database within a container database.

The static views are based on internal Oracle tables, such as `USER$`, `TAB$`, and `IND$`. If you have access to the `SYS` schema, you can view underlying tables directly via SQL.

The data dictionary tables (such as `USER$`, `TAB$`, `IND$`) are created during the execution of the `CREATE DATABASE` command. As part of creating a database, the `sql.bsq` file is executed, which builds these internal data dictionary tables. The `sql.bsq` file is generally located in the `ORACLE_HOME/rdbms/admin` directory; you can view it via an OS editing utility (such as `vi`, in Linux/Unix, or Notepad, in Windows).

The process of creating the static data dictionary views is:



You can view the creation scripts of the static views by querying the TEXT column of DBA_VIEWS:

```
SQL> set long 5000
SQL> select text from dba_views where view_name='DBA_VIEWS';
```

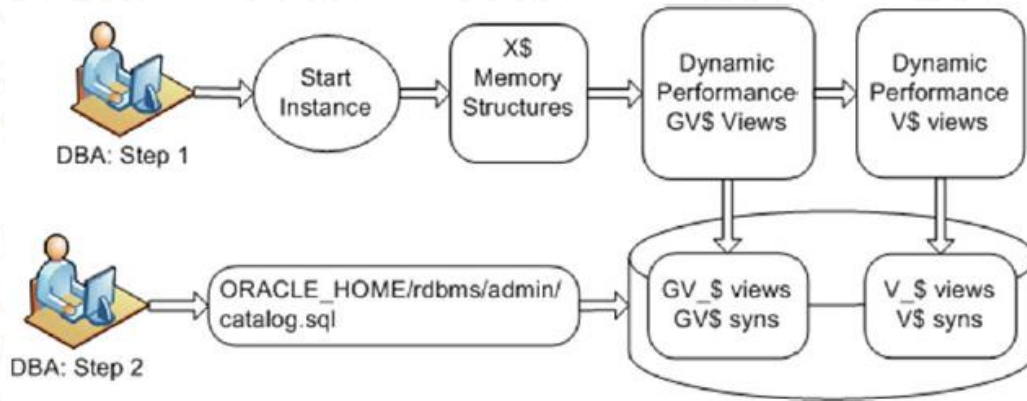
1.2. Dynamic Performance Views:

The dynamic performance data dictionary views are colloquially referred to as the v\$ and gv\$ views. These views are constantly updated by Oracle and reflect the current condition of the instance and database. Dynamic views are critical for diagnosing real-time performance issues.

The v\$ and gv\$ views are indirectly based on underlying x\$ tables, which are internal memory structures that are instantiated when you start your Oracle instance. Some of the v\$ views are available the moment the Oracle instance is started. For example, V\$PARAMETER contains meaningful data after the STARTUP NOMOUNT command has been issued, and does not require the database to be mounted or open. Other dynamic views (such as V\$CONTROLFILE) depend on information in the control file and therefore contain significant information only after the database has been mounted. Some v\$ views (such as V\$BH) provide kernel-processing information and thus have useful results only after the database has been opened.

At the top layer, the v\$ views are actually synonyms that point to underlying SYS.v_\$ views. At the next layer down, the SYS.v_\$ objects are views created on top of another layer of SYS.v\$ views. The SYS.v\$ views in turn are based on the SYS.gv\$ views. At the bottom layer, the SYS.gv\$ views are based on the x\$ memory structures. The top-level v\$ synonyms and SYS.v_\$ views are created when you run the catalog.sql script, which you usually do after the database is initially created.

The process of creating the Dynamic performance views is:



Accessing the v\$ views through the topmost synonyms is usually adequate for dynamic performance information needs. On rare occasions, you will want to query internal information that may not be available through the v\$ views.

If you work with Oracle RAC, you should be familiar with the gv\$ global views. These views provide global dynamic performance information regarding all instances in a cluster (whereas the v\$ views are instance specific). The gv\$ views contain an INST_ID column for identifying specific instances in a clustered environment. You can display the v\$ and gv\$ view definitions by querying the VIEW_DEFINITION column of the \$FIXED_VIEW_DEFINITION view:

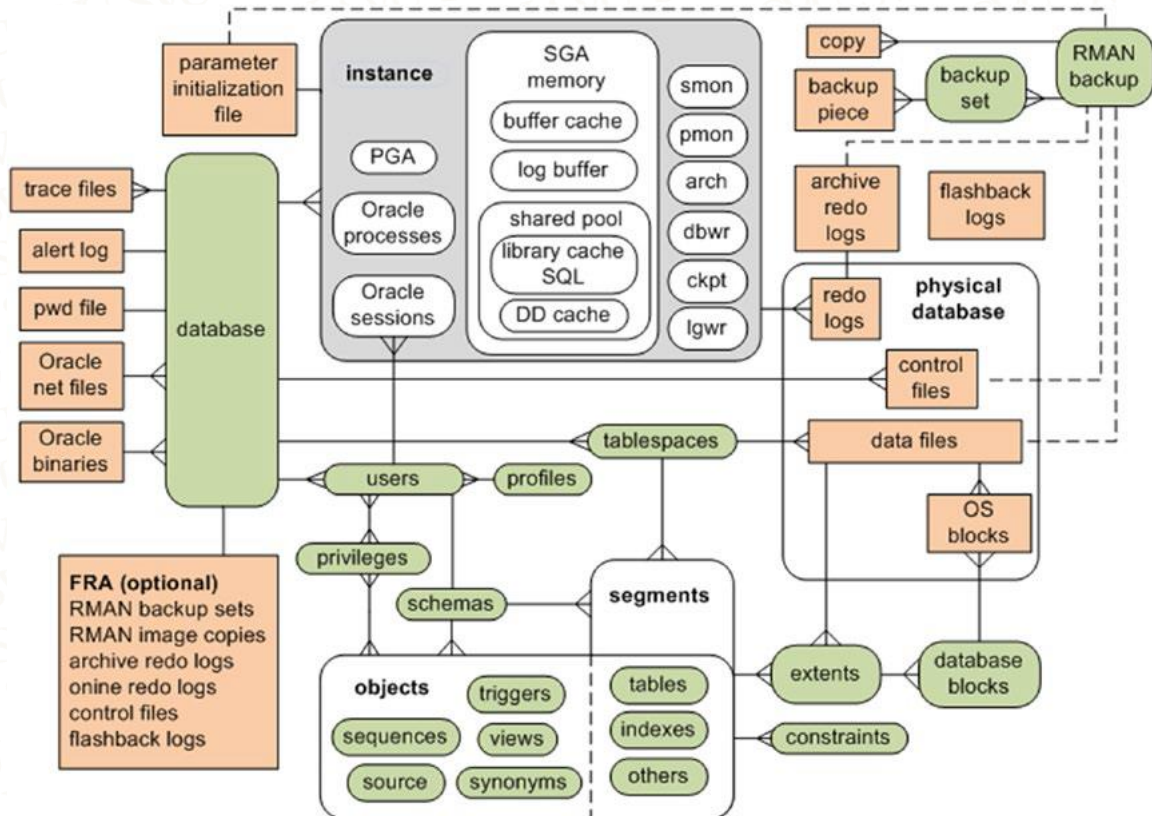
```
SQL> select view_definition
from v$fixed_view_definition
where view_name='V$CONTROLFILE';
```

2. A Different View of Metadata:

DBAs commonly face the following types of database issues:

- An insert into a table fails because a tablespace cannot extend
- The database is refusing connections because the maximum number of sessions is exceeded
- An application is hung, apparently because of some sort of locking issue
- A PL/SQL statement is failing, with a memory error
- RMAN backups have not succeeded for two days
- A user is trying to update a record, but a unique key constraint violation is thrown
- An SQL statement has been running for hours longer than normal
- Application users have reported that performance seems sluggish and that something must be wrong with the database

if a table cannot extend because a tablespace is full, what knowledge do you rely on to solve this problem? You need to understand that when a database is created, it contains multiple logical space containers called tablespaces. Each tablespace consists of one or more physical data files. Each data file consists of many OS blocks. Each table consists of a segment, and every segment contains one or more extents. As a segment needs space, it allocates additional extents within a physical data file. Once you understand the logical and physical concepts involved, you intuitively look in data dictionary views such as `DBA_TABLES`, `DBA_SEGMENTS`, `DBA_TABLESPACES`, and `DBA_DATA_FILES` to pinpoint the issue and add space as required. In a wide variety of troubleshooting scenarios, your understanding of the relationships of various logical and physical constructs will allow you to focus on querying views that will help you quickly resolve the problem at hand. The following data dictionary views map very closely to almost all the logical and physical elements of an Oracle database:



This diagram does provide you with a secure foundation on which to build your understanding of how to leverage the data dictionary views to get the data you need to do your job.

Note: There are several thousand CDB/DBA/ALL/USER static views and more than 700 v\$sql dynamic performance views.

3. A Few Creative Uses of the Data Dictionary:

Problem Solving using data dictionary

if you are troubleshooting an issue regarding materialized views, and you cannot remember the exact names of the data dictionary views associated with materialized views, you can do this:

```
SQL> select object_name
from dba_objects
where object_name like '%MV%'
and owner='SYS';
```

Often you need more information about each view. This is when the `DICTIONARY` and `DICT_COLUMNS` views can be invaluable. The `DICTIONARY` view stores the names of the data dictionary views. It has two columns:

```
SQL> desc dictionary
Name                Null?              Type
-----
TABLE_NAME          VARCHAR2 (30)
COMMENTS            VARCHAR2 (4000)
```

The previous query that gets the name of views related to materialized view can be written with dictionary:

```
SQL> select table_name, comments
from dictionary
where table_name like '%MV%';
```

--some lines of results

TABLE_NAME	COMMENTS
DBA_MVIEW_LOGS	All materialized view logs in the database
DBA_MVIEWS	All materialized views in the database
DBA_MVIEW_ANALYSIS	Description of the materialized views accessible to dba
DBA_MVIEW_COMMENTS	Comments on all materialized views in the database

If that does not give you enough information regarding the column names, you can query the `DICT_COLUMNS` view:

```
SQL> select column_name, comments
from dict_columns
where table_name='DBA_MVIEWS';
```

--Here is a fraction of the output:

COLUMN_NAME	COMMENTS
OWNER	Owner of the materialized view
MVIEW_NAME	Name of the materialized view
CONTAINER_NAME	Name of the materialized view container table
QUERY	The defining query that the materialized view Instantiates

Displaying User Information

You can run the following types of SQL commands to verify the currently connected user and database information:

```
SQL> show user;
SQL> select * from user_users;
SQL> select name from v$database;
SQL> select instance_name, host_name from v$instance;
```

You can also use the `SYS_CONTEXT` built-in SQL function to extract information from the data dictionary regarding details about your currently connected session:

```
SQL > select sys_context('USERENV','CURRENT_USER') usr
,sys_context('USERENV','AUTHENTICATION_METHOD') auth_mth
,sys_context('USERENV','HOST') host
,sys_context('USERENV','INSTANCE_NAME') inst
from dual;
```

Useful USERENV Parameters Available with SYS_CONTEXT:

Parameter Name	Description
AUTHENTICATED_IDENTITY	Identity used in authentication
AUTHENTICATION_METHOD	Method of authentication
CDB_NAME	Returns the name of the CDB; otherwise returns null
CLIENT_IDENTIFIER	Returns an identifier that is set by the application
CLIENT_INFO	User session information
CON_ID	Container identifier
CON_NAME	Container name
CURRENT_USER	Username for the currently active session
DB_NAME	Name specified by the DB_NAME initialization parameter
DB_UNIQUE_NAME	Name specified by the DB_UNIQUE_NAME initialization parameter
HOST	HOST Hostname for the machine on which the client initiated the database connection
INSTANCE_NAME	Instance name
IP_ADDRESS IP	IP address of the machine on which the client initiated the database connection
ISDBA	TRUE if the user authenticated with DBA privileges through the OS or password file

NLS_DATE_FORMAT	Date format for the session
OS_USER	OS user from the machine on which the client initiated the database connection
SERVER_HOST	Hostname of the machine on which the database instance is running
SERVICE_NAME	Service name for the connection
SID	Session identifier
TERMINAL	OS identifier for the client terminal

Displaying Table Row Counts

If you have accurate statistics, you can query the `NUM_ROWS` column of the `CDB/DBA/ALL/USER_TABLES` views. This column normally has a close row count if statistics are generated on a regular basis. The following query selects `NUM_ROWS` from the `USER_TABLES` view:

```
SQL> select table_name, num_rows from user_tables;
```

if you have partitioned tables and want to show row counts by partition, use the next few lines of SQL and PL/SQL code to generate the SQL required:

```
SQL> UNDEFINE user
SQL> SET SERVEROUT ON SIZE 1000000 VERIFY OFF
SQL> SPO part_count_&&user..txt
SQL> DECLARE
counter NUMBER;
sql_stmt VARCHAR2(1000);
CURSOR c1 IS SELECT table_name, partition_name FROM
dba_tab_partitions
WHERE table_owner = UPPER('&&user');
BEGIN
FOR r1 IN c1 LOOP
```

```
sql_stmt := 'SELECT COUNT(*) FROM &&user..' || r1.table_name || '
PARTITION ( ' || r1.partition_name || ' )';
EXECUTE IMMEDIATE sql_stmt INTO counter;
DBMS_OUTPUT.PUT_LINE(RPAD(r1.table_name
|| '(' || r1.partition_name || ')', 30) || ' ' || TO_CHAR(counter));
END LOOP;
END;
/
SPO OFF
```

You can generate statistics for all objects for a user with the following code:

```
SQL> exec dbms_stats.gather_schema_stats(ownname =>
'MV_MAINT', -
estimate_percent => DBMS_STATS.AUTO_SAMPLE_SIZE, -
degree => DBMS_STATS.AUTO_DEGREE, -
cascade => true);
```

The prior code instructs Oracle to estimate the percentage of the table to be sampled with the `ESTIMATE_PERCENT` parameter, using `DBMS_STATS.AUTO_SAMPLE_SIZE`. Oracle also chooses the appropriate degree of parallelism with the `DEGREE` parameter setting of `DBMS_STATS.AUTO_DEGREE`. The `CASCADE` parameter instructs Oracle to generate statistics for indexes.

Showing Primary Key and Foreign Key Relationships

It is useful to display data dictionary information regarding what primary key constraint is associated with a foreign key constraint. The following script queries the `DBA_CONSTRAINTS` data dictionary view to determine the parent primary key constraints that are related to child foreign key constraints:

```
SQL> select
a.constraint_type cons_type, a.table_name child_table
, a.constraint_name child_cons, b.table_name parent_table
, b.constraint_name parent_cons, b.constraint_type cons_type
from dba_constraints a, dba_constraints b
where a.owner = upper('&owner')
and a.table_name = upper('&table_name')
and a.constraint_type = 'R'
```

```
and a.r_owner = b.owner
and a.r_constraint_name = b.constraint_name;
```

The preceding script prompts you for two SQL*Plus ampersand variables (OWNER, TABLE_NAME);

When the CONSTRAINT_TYPE column (of DBA/ALL/USER_CONSTRAINTS) contains an R value, this indicates that the row describes a referential integrity constraint, which means that the child table constraint references a primary key constraint.

The next script takes the primary key record and looks to see if it has any child records with a constraint type of R. When you run this script, you are prompted for the primary key table owner and name:

```
SQL> select b.table_name primary_key_table, a.table_name
fk_child_table
,a.constraint_name fk_child_table_constraint
from dba_constraints a, dba_constraints b
where a.r_constraint_name = b.constraint_name
and a.r_owner = b.owner
and a.constraint_type = 'R'
and b.owner = upper('&table_owner')
and b.table_name = upper('&table_name');
```

Displaying Object Dependencies

Say you need to drop a table, but before you drop it, you want to display any objects that are dependent on it. Before making the change, you want to review what other objects are dependent on the table. You can use the DBA_DEPENDENCIES data dictionary view to display object dependencies. The following query prompts you for a username and an object name:

```
SQL> select '+' || lpad(' ',level+2) || type || ' ' || owner ||
'.' || name dep_tree
from dba_dependencies
connect by prior owner = referenced_owner and prior name =
referenced_name
and prior type = referenced_type
start with referenced_owner = upper('&object_owner')
and referenced_name = upper('&object_name')
and owner is not null;
```

The DUAL table

The DUAL table is part of the data dictionary. This table contains one row and one column and is useful when you want to return exactly one row, and you do not have to retrieve data from a particular table. In other words, you just want to return a value.

For example, you can perform arithmetic operations, as follows:

```
SQL> select 34*.15 from dual;  
34*.15  
-----  
5.1
```

Other common uses are selecting from DUAL to show the current date or to display some text within an SQL script.

4. Quiz:

1. Which of these statements about data dictionary is false?
 - a) The data dictionary stores critical information about the physical characteristics of the database
 - b) The data dictionary stores critical information about the users
 - c) The data dictionary stores critical information about the objects
 - d) The data dictionary stores critical information about the operating system hotspots
2. Is used by the Oracle Database server to find information about users, objects, constraints, and storage
 - a) System privileges.
 - b) Roles
 - c) Object privileges
 - d) Data dictionary
3. Which of these statements is incorrect regarding Data Dictionary? (Choose the best answer).
 - a) Is available for use by any user to query information about the database
 - b) Is owned by the SYS user
 - c) Should be modified directly using SQL
 - d) Should never be modified directly using SQL
4. Static views of data dictionary will have information about Changes to your database, such as:
 - a) SQL currently executing
 - b) Adding a user
 - c) Memory usage
 - d) Locks

5. The data dictionary stores critical information about the physical characteristics of the database, users, objects, and dynamic performance metrics.
- a) True
 - b) False
6. Dynamic performance Views of data dictionary will have information about Changes to your database, such as:
- a) Adding a user
 - b) Memory usage
 - c) Creating a table
 - d) Modifying a column
7. Dynamic performance Views of data dictionary sometimes called:
- a) The USER_ Views
 - b) The v\$ or GV\$ Views
 - c) The DBA_Views
 - d) The All_Views
8. Static Views of data dictionary sometimes called:
- a) The v\$ Views
 - b) The GV\$ Views
 - c) The USER/ALL/DBA/CDB_Views
 - d) The v_\$ Views
9. USER_TABLES view contains information about tables owned by the current user, is it?
- a) From Static Views
 - b) From Dynamic Performance Views
 - c) From Both
 - d) Doesn't belong to Data Dictionary

10. ALL_TABLES view contains information about tables accessed by the current user, is it?

- a) From Dynamic Performance Views
- b) From Static Views
- c) From Both
- d) Doesn't belong to Data Dictionary

11. DBA_TABLES view contains information about all tables of the database, is it?

- a) From Dynamic Performance Views
- b) From Both
- c) From Static Views
- d) Doesn't belong to Data Dictionary

12. CDB_TABLES view contains information about all tables of the pluggable database, is it?

- a) From Dynamic Performance Views
- b) From Both
- c) Doesn't belong to Data Dictionary
- d) From Static Views

13. V\$Parameter view contains information about the initialization parameters that are currently in effect for the session, is it?

- a) From Dynamic Performance Views
- b) From Both
- c) Doesn't belong to Data Dictionary
- d) From Static Views

14. To access the DBA views, the current user must be granted the roles?
- a) A CONNECT role and RESOURCE role
 - b) A DBA role or SELECT_CATALOG_ROLE
 - c) A CONNECT role or RESOURCE role
 - d) A AUDIT_ADMIN and AQ_USER_ROLE
15. What is the View that contains the names of the data dictionary views?
- a) DICTIONARY
 - b) DBA_MVIEW_LOGS
 - c) V\$PARAMETER
 - d) V\$LOCK
16. What is the View that contains the names of the data dictionary views and their column names?
- a) DICTIONARY
 - b) DICT_COLUMNS
 - c) V\$PARAMETER
 - d) V\$LOCK
17. Why do you use the SYS_CONTEXT built-in SQL function?
- a) To extract information from the data dictionary regarding Memory structures
 - b) To extract information from the data dictionary regarding database files
 - c) To extract information from the data dictionary regarding details about your currently connected session
 - d) To extract general information from the data dictionary
18. You can use the DBA_DEPENDENCIES data dictionary view to:
- a) Display SYSTEM object only
 - b) Display SYS object only
 - c) Display constraint references
 - d) Display object dependencies

19. The DUAL table is not a part of the data dictionary?

- a) True
- b) False

20. The data dictionary tables (such as USER\$, TAB\$, IND\$) are created during the execution of the CREATE DATABASE command?

- a) True
- b) False

Quiz answers – chapter 10	
1	d
2	d
3	c
4	b
5	a
6	b
7	b
8	c
9	a
10	b
11	c
12	d
13	a
14	b
15	a
16	b
17	c
18	d
19	b
20	a

References:

Books

1. Oracle Database Administrator's Guide, 18c (Oracle University), March 2019, Primary Author: Rajesh Bhatiya, Randy Urbano
2. database-new-features-guide (Oracle University), 18c E88909-01 February 2018, Primary Authors: Tanaya Bhattacharjee, Sunil Surabhi, Mark Baue
3. The Cloud DBA-Oracle: Managing Oracle Database in the Cloud – First Edition, Abhinivesh Jain and Niraj Mahajan. Apress – ISBN-13 (pbk): 978-1-4842-2634-6 ISBN-13 (electronic): 978-1-4842-2635-3

Web Sites

1. <https://docs.oracle.com/database>
2. <https://docs.oracle.com/en/database/oracle/oracle-database/19/whats-new.html>
3. <http://www.oracle.com/technetwork/database/enterprise-edition/downloads/index.html>